

Lecture 3

–

NumPy

Buntspecht / Great Spotted Woodpecker

(*Dendrocopos major, Syn.: Picoides major*)

- Medium-sized woodpecker with pied black and white plumage and a red patch on the lower belly. Males and young birds also have red markings on the neck or head.
- Great spotted woodpeckers chisel into trees to find food or excavate nest holes, and also drum for contact and territorial advertisement.
- Like other woodpeckers, they have anatomical adaptations to manage the physical stresses from the hammering action.
- The call of the great spotted woodpecker is a sharp *kik*, which may be repeated as a wooden rattling *krrarraarr* if the bird is disturbed.
- The great spotted woodpecker spends much of its time climbing trees.

Sources:
* Photo by Josef Gadermaier on pixabay
* https://en.wikipedia.org/wiki/Great_spotted_woodpecker

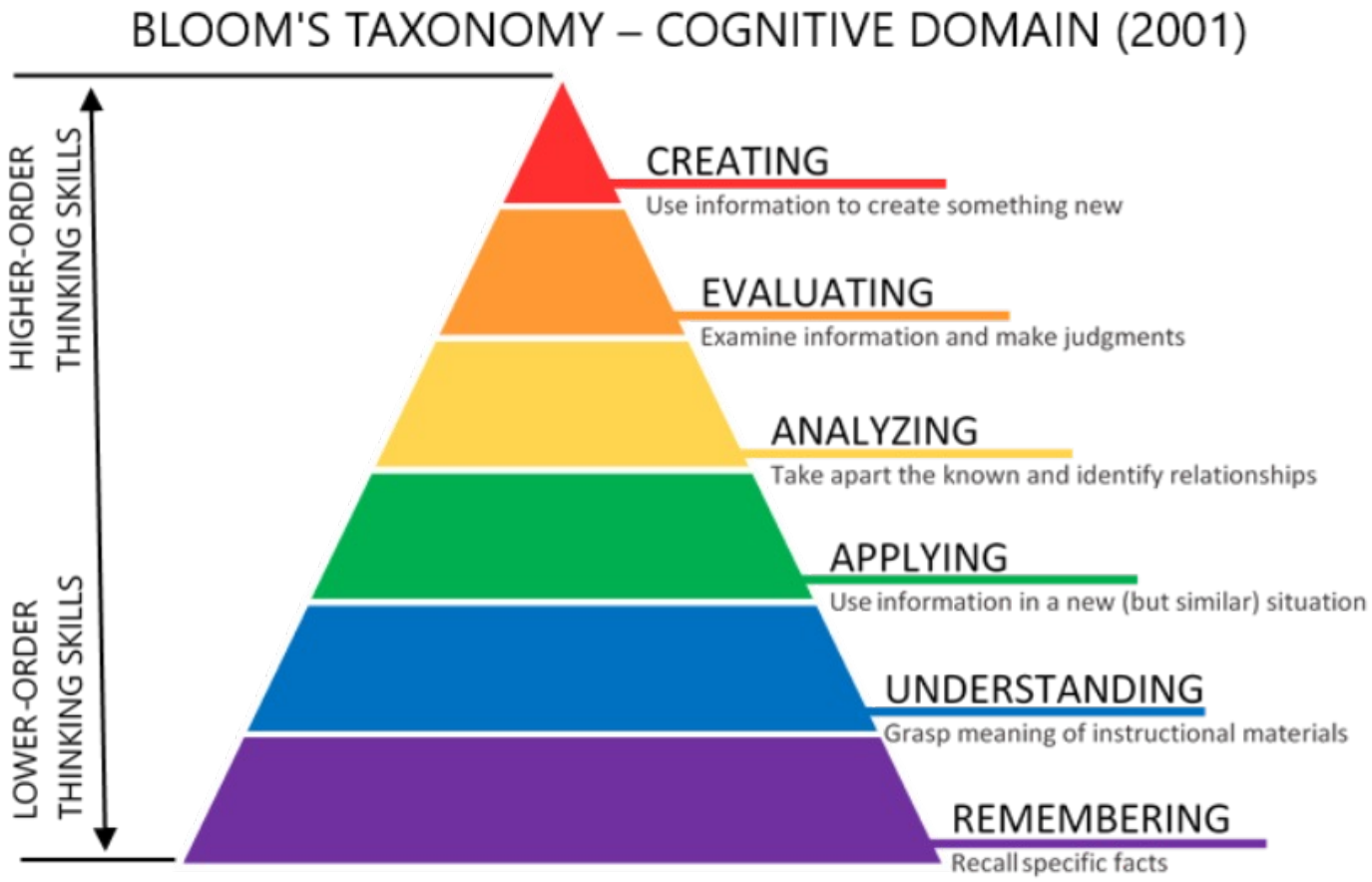
Focusing Objectives

By the end of today's lecture, you will be able to:

1. Understand the Role of NumPy as the Basis of more advanced Libraries
2. Work with NumPy Arrays and Data Types
3. Apply Indexing and Slicing Techniques
4. Perform Array Math and Use Broadcasting



Motivation



Source: https://citt.ufl.edu/resources/the-learning-process/designing-the-learning-experience/blooms-taxonomy/

Data Science

NumPy

1. Library Overview
2. Arrays & Indexing
3. Array Math & Broadcasting

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Important Python Libraries for Data Science













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NumPy - *"The fundamental package for scientific computing with Python"*

- POWERFUL N-DIMENSIONAL ARRAYS
  - Fast and versatile, the NumPy vectorization, indexing, and broadcasting concepts are the de-facto standards of array computing today.
- NUMERICAL COMPUTING TOOLS
  - NumPy offers comprehensive mathematical functions, random number generators, linear algebra routines, Fourier transforms, and more.
- PERFORMANT
  - The core of NumPy is well-optimized C code. Enjoy the flexibility of Python with the speed of compiled code.
- EASY TO USE
  - NumPy's high level syntax makes it accessible and productive for programmers from any background or experience level.



Full Documentation:
https://numpy.org/

Source: https://numpy.org/

ndarray and dtype

- Provides n-dimensional array object & routines for fast operations on ndarrays
  - all elements in the array required to be of the same data type
  - fixed size at creation
  - most operations performed in compiled C code for performance
- NumPy's array class is called `ndarray` (alias `array`)
  - Note: `numpy.array` is not the same as the Standard Python Library class `array.array`!
- Important attributes of `ndarray` objects:
  - `ndarray.ndim`: number of axes (dimensions) of the array.
  - `ndarray.shape`: dimensions of the array: tuple of integers indicating the size of the array in each dimension.
    - For a matrix with *n* rows and *m* columns, shape will be (n,m).
    - Length of the shape tuple is therefore the number of axes (ndim).
  - `ndarray.dtype`: object describing the type of the elements in the array, e.g., `numpy.int64`, `numpy.float64`

NumPy

1. Library Overview
2. Arrays & Indexing
3. Array Math & Broadcasting

Demo

Demo

NumPy Arrays and Indexing



Source: Foto von Markus Spiske auf Unsplash

Exercise

Exercise 1

NumPy - Part 1

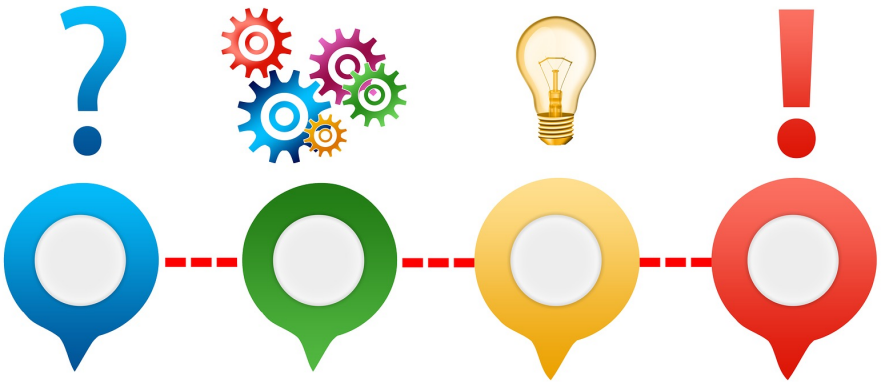


Photo by Gerd Altmann on Pixabay

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Demo

Demo

NumPy Array Math and Broadcasting



Source: Foto von Markus Spiske auf Unsplash

Exercise

Exercise 2

NumPy – Part 2

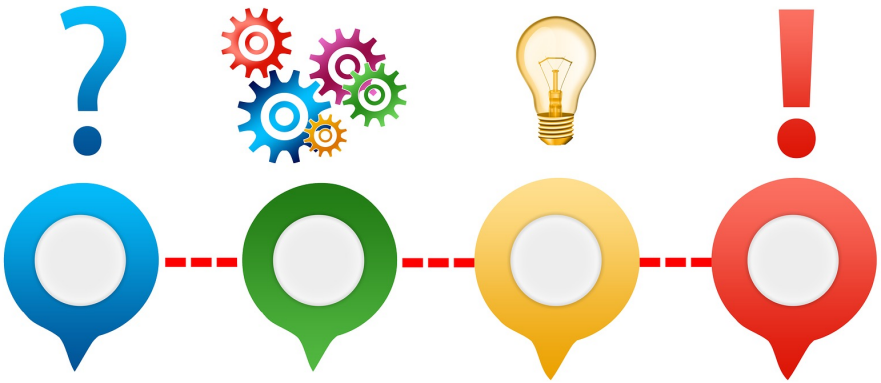


Photo by Gerd Altmann on Pixabay

Key Takeaways

- Python Libraries for Data Science
  - NumPy, pandas, Matplotlib, seaborn, scikit-learn
- NumPy data structure (nd)array
  - Creating ndarrays
  - Indexing ndarrays, specifically integer and Boolean indexing
  - Math operations on ndarrays
  - Broadcasting



Image by Gordon Johnson on pixabay